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Course: Calculus Beta

Assignment: MyLab Math - Uge 9

If $f(x) = kx^2$, determine the value of k that makes $f(x)$ a probability density function on $0 \leq x \leq 11$.

Recall that a function $f(x)$ defined on an interval $A \leq x \leq B$ is a probability density function if it satisfies the following properties.

(I) $f(x) \geq 0$ for $A \leq x \leq B$.

(II) $\int_A^B f(x) dx = 1$.

Let $f(x) = kx^2$. Note that if a is any number in the interval $0 \leq x \leq 11$, then a^2 is a non-negative number. Thus, ka^2 must be non-negative as well in order to satisfy the first property of probability density functions. In other words, k cannot be negative.

By the second property of probability density functions, k must satisfy the equation

$$\int_0^{11} kx^2 dx = 1$$

Evaluate the integral $\int_0^{11} kx^2 dx$.

$$\begin{aligned} \int_0^{11} kx^2 dx &= k \int_0^{11} x^2 dx \\ &= k \left[\frac{x^3}{3} \right]_0^{11} \end{aligned}$$

Thus,

$$\begin{aligned} 1 &= k \left[\frac{x^3}{3} \right]_0^{11} \\ &= k \left[\frac{11^3}{3} - \frac{0^3}{3} \right] \\ &= k \left(\frac{1331}{3} \right) \end{aligned}$$

Therefore, the value of k that makes $f(x) = kx^2$ a probability density function is $\frac{3}{1331}$.